

Marble Solver Derivation

Robotic Exploration Lab

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1 Problem Setup

We seek to solve the following Quadratic Program with Complementarity Constraints (QPCC):

$$\begin{aligned} & \underset{x \in \mathbb{R}^n}{\text{minimize}} && \frac{1}{2}x^\top Qx + g^\top x + c \\ & \text{s.t.} && Ax + b = 0 \\ & && Gx + h \geq 0 \\ & && 0 \leq Lx + l \perp Rx + r \geq 0 \end{aligned} \tag{1}$$

for $Q \succeq 0$, $A \in \mathbb{R}^{m \times n}$, $G \in \mathbb{R}^{p \times n}$, $L, R \in \mathbb{R}^{c \times n}$, $b \in \mathbb{R}^m$, $h \in \mathbb{R}^p$, and $l, r \in \mathbb{R}^c$.

2 KKT System Derivation

We introduce the *retraction map* $b_\kappa(x)$ which satisfies:

$$b_\kappa(x) \cdot b_\kappa(-x) = \kappa \tag{2}$$

Using (2), we can relax the complementarity constraints in (1) as:

$$\begin{aligned} & \underset{x \in \mathbb{R}^n, \sigma \in \mathbb{R}^c, d \in \mathbb{R}^p}{\text{minimize}} && \frac{1}{2}x^\top Qx + g^\top x + c - \kappa \mathbf{1}^T \log(d) \\ & \text{s.t.} && Ax + b = 0 \\ & && Gx + h = d \\ & && Lx + l = b_\kappa(\sigma) \\ & && Rx + r = b_\kappa(-\sigma) \end{aligned} \tag{3}$$

We then introduce the Augmented Lagrangian for (3) as:

$$\begin{aligned} \mathcal{L}_\rho(x, \sigma, d; \alpha, \beta, \gamma_L, \gamma_R) = & \frac{1}{2}x^\top Qx + g^\top x + c - \kappa \mathbf{1}^T \log(d) \\ & + \alpha^\top (Ax + b) + \frac{\rho}{2} \|Ax + b\|^2 \\ & + \beta^\top (Gx + h - d) + \frac{\rho}{2} \|Gx + h - d\|^2 \\ & + \gamma_L^\top (Lx + l - b_\kappa(\sigma)) + \frac{\rho}{2} \|Lx + l - b_\kappa(\sigma)\|^2 \\ & + \gamma_R^\top (Rx + r - b_\kappa(-\sigma)) + \frac{\rho}{2} \|Rx + r - b_\kappa(-\sigma)\|^2 \end{aligned} \tag{4}$$

Using an Augmented-Lagrangian approach, we can solve (3) by iteratively minimizing (4) over primal variables x, σ, d .

We first derive stationarity with respect to the original primal variable x . Define the following shorthands:

$$r_A := Ax + b \quad (5)$$

$$r_G := Gx + h - d \quad (6)$$

$$r_L := Lx + l - b_\kappa(\sigma) \quad (7)$$

$$r_R := Rx + r - b_\kappa(-\sigma) \quad (8)$$

We therefore have the following stationarity condition:

$$\frac{\partial \mathcal{L}_\rho}{\partial x} = Qx + g + A^\top \alpha + \rho A^\top r_A + G^\top \beta + \rho G^\top r_G + L^\top \gamma_L + \rho L^\top r_L + R^\top \gamma_R + \rho R^\top r_R = 0 \quad (9)$$

$$= Qx + g + A^\top \lambda + G^\top \mu + L^\top \tau_L + R^\top \tau_R = 0 \quad (10)$$

We introduce the primal-dual shifted multiplier variables:

$$\lambda = \alpha + \rho r_A = \alpha + \rho(Ax + b) \quad (11)$$

$$\mu = \beta + \rho r_G = \beta + \rho(Gx + h - d) \quad (12)$$

$$\tau_L = \gamma_L + \rho r_L = \gamma_L + \rho(Lx + l - b_\kappa(\sigma)) \quad (13)$$

$$\tau_R = \gamma_R + \rho r_R = \gamma_R + \rho(Rx + r - b_\kappa(-\sigma)) \quad (14)$$

For variable d , we have the following stationarity condition:

$$\frac{\partial \mathcal{L}_\rho}{\partial d} = -\mu - \kappa d^{-1} = 0 \quad (15)$$

$$\iff \mu \odot d = -\kappa \mathbf{1} \quad (16)$$

We can satisfy this relaxed complementarity condition by again using (2) as:

$$d = b_\kappa(v), \quad \mu = -b_\kappa(-v), \quad v \in \mathbb{R}^p \quad (17)$$

And finally for variable σ , we have the following stationarity condition:

$$\frac{\partial \mathcal{L}_\rho}{\partial \sigma} = -b'_\kappa(\sigma)(\gamma_L + \rho r_L) + b'_\kappa(-\sigma)(\gamma_R + \rho r_R) = 0 \quad (18)$$

$$= -b'_\kappa(\sigma)\tau_L + b'_\kappa(-\sigma)\tau_R = 0 \quad (19)$$

In all, we have the following KKT conditions for (3):

$$r_x := Qx + g + A^\top \lambda + G^\top \mu + L^\top \tau_L + R^\top \tau_R = 0 \quad (20)$$

$$r_v := -\mu - b_\kappa(-v) = 0 \quad (21)$$

$$r_\sigma := -b'_\kappa(\sigma)\tau_L + b'_\kappa(-\sigma)\tau_R = 0 \quad (22)$$

$$r_\lambda := Ax + b - \rho^{-1}(\lambda - \alpha) = 0 \quad (23)$$

$$r_\mu := Gx + h - b_\kappa(v) - \rho^{-1}(\mu - \beta) = 0 \quad (24)$$

$$r_{\tau_L} := Lx + l - b_\kappa(\sigma) - \rho^{-1}(\tau_L - \gamma_L) = 0 \quad (25)$$

$$r_{\tau_R} := Rx + r - b_\kappa(-\sigma) - \rho^{-1}(\tau_R - \gamma_R) = 0 \quad (26)$$

We now write the asymmetric Newton system for the KKT conditions described above.

$$\begin{bmatrix} Q & 0 & 0 & A^\top & G^\top & L^\top & R^\top \\ 0 & B_v^- & 0 & 0 & -I & 0 & 0 \\ 0 & 0 & \text{diag}(-b''_\kappa(\sigma) \odot \tau_L - b''_\kappa(-\sigma) \odot \tau_R) & 0 & 0 & -B_\sigma^+ & B_\sigma^- \\ A & 0 & 0 & -\rho^{-1}I & 0 & 0 & 0 \\ G & -B_v^+ & 0 & 0 & -\rho^{-1}I & 0 & 0 \\ L & 0 & -B_\sigma^+ & 0 & 0 & -\rho^{-1}I & 0 \\ R & 0 & B_\sigma^- & 0 & 0 & 0 & -\rho^{-1}I \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta v \\ \Delta \sigma \\ \Delta \lambda \\ \Delta \mu \\ \Delta \tau_L \\ \Delta \tau_R \end{bmatrix} = - \begin{bmatrix} r_x \\ r_v \\ r_\sigma \\ r_\lambda \\ r_\mu \\ r_{\tau_L} \\ r_{\tau_R} \end{bmatrix} \quad (27)$$

with the following Jacobian blocks:

$$B_v^+ := b'_\kappa(v) \quad (28a)$$

$$B_v^- := b'_\kappa(-v) \quad (28b)$$

$$B_\sigma^+ := b'_\kappa(\sigma) \quad (28c)$$

$$B_\sigma^- := b'_\kappa(-\sigma) \quad (28d)$$

2.1 Symmetrization 1

We simply scale the second row of the Newton system by B_v^+ to obtain the following symmetric Newton system:

$$\begin{bmatrix} Q & 0 & 0 & A^\top & G^\top & L^\top & R^\top \\ 0 & B_v^+ B_v^- & 0 & 0 & -B_v^+ & 0 & 0 \\ 0 & 0 & \text{diag}(-b''_\kappa(\sigma) \odot \tau_L - b''_\kappa(-\sigma) \odot \tau_R) & 0 & 0 & -B_\sigma^+ & B_\sigma^- \\ A & 0 & 0 & -\rho^{-1}I & 0 & 0 & 0 \\ G & -B_v^+ & 0 & 0 & -\rho^{-1}I & 0 & 0 \\ L & 0 & -B_\sigma^+ & 0 & 0 & -\rho^{-1}I & 0 \\ R & 0 & B_\sigma^- & 0 & 0 & 0 & -\rho^{-1}I \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta v \\ \Delta \sigma \\ \Delta \lambda \\ \Delta \mu \\ \Delta \tau_L \\ \Delta \tau_R \end{bmatrix} = - \begin{bmatrix} r_x \\ B_v^+ r_v \\ r_\sigma \\ r_\lambda \\ r_\mu \\ r_{\tau_L} \\ r_{\tau_R} \end{bmatrix} \quad (29)$$

2.2 Symmetrization 2

We symmetrize the Newton system by introducing the diagonal matrix $W := (B_v^+)^{1/2}$ and introducing auxiliary variable $\Delta v = W^{-1} \Delta \tilde{v}$.

$$\begin{bmatrix} Q & 0 & 0 & A^\top & G^\top & L^\top & R^\top \\ 0 & B_v^- & 0 & 0 & -W & 0 & 0 \\ 0 & 0 & \text{diag}(-b''_\kappa(\sigma) \odot \tau_L - b''_\kappa(-\sigma) \odot \tau_R) & 0 & 0 & -B_\sigma^+ & B_\sigma^- \\ A & 0 & 0 & -\rho^{-1}I & 0 & 0 & 0 \\ G & -W & 0 & 0 & -\rho^{-1}I & 0 & 0 \\ L & 0 & -B_\sigma^+ & 0 & 0 & -\rho^{-1}I & 0 \\ R & 0 & B_\sigma^- & 0 & 0 & 0 & -\rho^{-1}I \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta \tilde{v} \\ \Delta \sigma \\ \Delta \lambda \\ \Delta \mu \\ \Delta \tau_L \\ \Delta \tau_R \end{bmatrix} = - \begin{bmatrix} r_x \\ -W r_v \\ r_\sigma \\ r_\lambda \\ r_\mu \\ r_{\tau_L} \\ r_{\tau_R} \end{bmatrix} \quad (30)$$

and we recover $\Delta v = W^{-1} \Delta \tilde{v}$ after solving the system.